

Ontario Provincial Police

Ammunition

Purchase Requirement

PR-8310-OPP-2TF

1. ITEM: .223 Rem, Frangible Ammunition (Training)

2. GENERAL REQUIRMENTS:

- a. Shall meet all requirements detailed in PR-8310-OPP-ALL

3. SPECIFIC REQUIREMENTS:

- a. Manufacturer shall complete Attachment A – Manufacture Data Sheet and return to O.P.P. with any submissions
- b. Bullet pull force will be sufficient so as not to dislodge the projectile from its seated position in a previously un-chambered cartridge during recoil in O.P.P. firearms or their magazines
- c. The fired projectile shall not penetrate currently issued O.P.P. NIJ Level III (plus special threats) armour plates
 - i. Plate A
 - 1. Manufacturer HESCO ARMOR LLC
 - 2. Model # AMUR 3620
 - ii. Plate B
 - 1. Manufacturer DEW ENGINEERING (Formally BLS TEXTILES INC)
 - 2. Model # 5004
- d. Projectile must be frangible construction and be lead free.
- e. Cartridge case shall be manufactured from brass and shall not be nickel plated.
- f. It is preferred the cartridge case mouth and primer be sealed to prevent the entrance of moisture, which could alter the performance of components in various environmental conditions.
- g. Primer shall be lead-free and be non-corrosive.

h. Frangibility

Although not developed for rifle ammunition, ammunition will be tested and shall meet the requirements for frangibility as published in SAAMI-Z299.3 (2022) Voluntary Industry Performance Standards for Pressure and Velocity of Centerfire Pistol and Revolver Ammunition for the Use of Commercial Manufacturers

4. Packaging and identification

- a. Ammunition may be contained in cartons of 20 or 50 rounds per carton and cases of 200, 500 or 1,000 rounds.

5. Quality Control Plan/Quality Assurance Program

- a. See PR-8310-OPP-ALL for details and Deliverables 1.5.6 Vendor Quality Control Plan and Quality Control Program.

6. Essential Quality Standards

- a. VELOCITY – An average of 10 at 15ft: +/- 90 f/s from SAAMI published velocity for projectile weight when tested from a SAAMI 24” test barrel. Based on ANSI/SAAMI Z299.4-2015
- b. PRESSURE – A maximum average of 10: 55,000 P.S.I. using piezoelectric system for pressure measuring and SAAMI test equipment. Based on ANSI/SAAMI Z299.4-2015

PRIMER – Heavy metal-free (lead free) primer must be used, it is preferred that the primer be sealed

- c. DEFECTIVE TOLERANCES – Ammunition not falling within the noted tolerances may be subject to return to the supplier together with all the ammunition from the same lot for replacement or monetary return at the discretion of the O.P.P. Ammunition returned under this condition will be at the supplier’s cost.

ANSI Standard on Inspection by Attributes ANSI/ASQ Z-1.4-2003 (R2013), Table I-Sample size code letters, General inspection level II, Table III-A Double sampling plans for normal inspection (Master table).

DEFECT DESCRIPTION	ALLOWABLE QUALITY LEVEL
Critical (Functional)	0.00

Any ammunition manufacturing defect which would result in cartridge failure producing a firearms blow-up. Any loading or propellant powder defect which may result in a bullet lodging in the bore of a firearm.

DEFECT DESCRIPTION	ALLOWABLE QUALITY LEVEL
Major (Functional)	0.040

Misfires, hard extractions or any other defect which would seriously alter functioning or performance.

DEFECT DESCRIPTION	ALLOWABLE QUALITY LEVEL
Major (Visual & Dimensional)	0.040

Any visual or dimensional defect which would seriously alter functioning or performance.

DEFECT DESCRIPTION	ALLOWABLE QUALITY LEVEL
Minor (Visual & Dimensional)	0.25

Any visual or dimensional defect which would not seriously alter functioning or performance.

7. Performance Standards

a. Accuracy Specification

Five – 5 shot groups must average not more than 15.24 cm (6”) at 100m (109.36 yds) fired from a Colt Canada C8 CQB with 25.4 cm (10”) barrel and Vortex flash suppressor, hand held on a sand bag rest.

and/or

A 25 shot group fired from a universal receiver mounted accuracy test barrel manufactured to SAAMI standards for a .223 Remington caliber shall be less than 10.16 cm (4”) at 100m (109.36 yds).

Ammunition not meeting this accuracy specification will be treated as exceeding the defect tolerance.

- b. Ammunition must be compatible (feed and function) with the following firearms
 - i. Colt Canada C8 CQB with 25.4 cm (10”) barrel and Vortex flash suppressor

- ii. Colt Canada C8 MRR with 29.5 cm (11.6") barrel and vortex flash suppressor
- iii. Colt Canada C8 A2 with 35.56 cm (14") barrel and a slotted military style flash hider or a Vortex flash suppressor

8. Identifications Standards

- a. See PR-8310-OPP-ALL for details with addition(s) of individual specifications detailed in Para 4 of individual specifications.

9. Quality Assurance

- a. See PR-8310-OPP-ALL for details

ATTACHMENT A

Manufacture Data Sheet

.223 Rem, Frangible Ammunition (Training)

Manufacturer:			
Principal Technical Contact:			
Telephone:			
Email address:			
Caliber:	.223 Remington		
Product name:		Purpose:	Training
Product Order/Part #:		Sub-Purpose:	Frangible
Rounds per box:			
Boxes per case:			
Projectile composition:			
Ballistic Coefficient:			
Sectional Density:			
LOT/DAY CODE numbers:	Lot 1:		Lot 2:
Meaning of characters in LOT/DAY CODE numbers:			

Component compositions				
ITEM	MANUFACTURER	TYPE	WEIGHT (if applicable)	
Projectile:				
Case mouth Sealant:				
Case:				
Primer:				
Primer Sealant:				
Propellant:				
Headstamp marking:				
Headstamp meaning:				
Material Safety Data sheets attached:				
Production group size standard measured at 100m (109.36 yds):		Cm (Inches)	# of rounds fired:	
Recommended velocity window for optimal terminal performance in ballistic gelatin (fps):	Minimum:		Maximum:	

Exterior ballistics data based on manufacturer's averages— C8 COB , 100 m zero

Distance	Velocity (fps)	Energy (foot pounds)	Trajectory (inches)	
Muzzle				
5				
10				
15				
20				
25				
50				
Maximum Range		Yards @		Angle